



terms of compactness, robustness, reliability and logistic simplicity. However, with the present technology status, it cannot match the output laser power of megawatt level achievable in chemical lasers.

Power level of tens of kilowatts have already been realised in bulk solid-state lasers. The present technology thrust is to realise 100-kilowatt-class solid-state laser source, which would be a good enough power level for a short-range tactical directed-energy laser weapon for counter-UAV and counter-RAM applications.

Salient features of solid-state lasers of relevance to directed-energy laser weapon development include shorter wavelength, absence of chemical hazards and higher efficiency. Shorter wavelength of one-micron yields lower divergence for a given transmitting aperture size and hence longer operating ranges. Also, shorter wavelength offers better laser-matter interaction. All-solid-state electrically-operated configuration leads to a lightweight, compact and rugged system without any chemical hazards and requirement of any special support logistics. High efficiency means low operating cost and a deep magazine.

Research and development efforts in the recent years at Lawrence Livermore National Laboratory, USA, and Northrop Grumman, USA, have generated a lot of interest in high-power solid-state lasers internationally for futuristic laser based DEWs.

High-power solid-state laser programmes being pursued at the two centres follow two different design architectures. Both have the objective of developing a 100-kilowatt-class single-aperture solid-state laser. While Lawrence Livermore National Laboratory is developing heat capacity disk laser technologies, Northrop Grumman's Joint High Power Solid State Laser (JHPSSL)

programme (Fig. 9) is based on oscillator-amplifier configuration.

Another high-power solid-state laser based on the slab architecture previously used by Northrop Grumman to build 105-kilowatt laser is Gamma laser. The company announced Gamma in 2012 after completing an extensive series of initial tests. The laser operated at 13.3 kilowatts for a number of shots over a total of 1.5 weeks with stable performance and a beam quality that exceeded design goals, completing the initial phase of trials.



Fig. 9: Solid-state laser developed under JHPSSL programme

Gamma laser is the basic building block that is designed in such a way that more than one such laser could be combined to produce higher power levels. The laser has also been ruggedised for survivability in real-world operational environment by subjecting key portions of the laser to vibration, shock and thermal testing.

Fibre lasers

The technology of fibre lasers is the most advanced solid-state laser technology available today. The basic fibre laser comprises a gain medium in the form of a long optical fibre of suitable material doped with laser ions. For high-power operation, Ytterbium-doped glass fibre is typically used. The entire fibre length is pumped with a large number of single-emitter fibre coupled laser diode arrays.

The laser resonator cavity is formed by embedding Bragg grating reflectors at the two ends of the fibre. Due to the small aperture of fibre (few microns), output laser beam is emitted with diffraction-limited beam quality, resulting in output laser intensity (brightness), which is nearly two orders of magnitude higher as compared to that produced by conventional solid-state lasers.

Also, since the resonator is formed within the fibre, there is no free space optics, thereby making fibre laser extremely robust and reliable as compared to other lasers. Salient features of fibre lasers are summarised as follows:

Shorter wavelength of one micron offers lower beam divergence and therefore longer operating ranges. Shorter wavelength also has better laser-matter interaction. Fibre lasers produce diffraction-limited beam quality. These are relatively more efficient with wall-plug efficiency of about 25 per cent as compared to 10 per cent in the case of diode-pumped bulk solid-state lasers. These are also more reliable and robust due to absence of free-space optics.

The technology of fibre laser is extremely complex and confined to only couple of international companies. Single-mode fibre lasers with an output power of 400 watts - 600 watts are commercially available. The technology of kilowatt-class fibre laser is available with only a few companies including SPI Lasers UK Ltd, UK, and IPG Photonics, USA.

SPI lasers offers single-mode fibre lasers up to one kilowatt output power. At IPG Photonics, single-mode fibre lasers up to ten kilowatt power level and a beam quality characterised by M^2 -value better than 1.07 are well established. Multi-mode fibre lasers up to 50 kilowatt power are also available. Higher single-mode output power levels are realisable by using coherent combinations of multiple fibre laser beams.

To be continued...